1 **(a)** Scale factor $= \dfrac{18}{6} = 3$ **[M1]**

15 **[A1]**

(b) Area scale factor $= 3^2 = 9$ **[M1]**

$\dfrac{72}{9} = 8$ **[A1]**

[Total Marks: 4]

2 **(a)** Volume scale factor $= \left(\dfrac{15}{6}\right)^3 = \dfrac{125}{8}$ **[M1]**

$V_B = 48 \times \dfrac{125}{8} = 750$ **[A1]**

(b) Area scale factor $= \left(\dfrac{15}{6}\right)^2 = \dfrac{25}{4}$ **[M1]**

$A_A = 200 \div \dfrac{25}{4} = 32$ **[A1]**

[Total Marks: 4]

3 **(a)** Angle A is common to both triangles **[B1]**

Angle ABC = Angle ADE (corresponding angles as BC is parallel to DE) **[B1]**

(b) $\dfrac{DE}{7.5} = \dfrac{6+4}{6}$ **[M1]**

12.5 **[A1]**

(c) Area scale factor $= \left(\dfrac{10}{6}\right)^2 = \dfrac{25}{9}$ **[M1]**

Area of $ADE = 18 \times \dfrac{25}{9} = 50$ **[M1]**

Area of $BCED = 50 - 18 = 32$ **[A1]**

[Total Marks: 7]

4 **(a)** Volume ratio $A : B = \dfrac{108}{500} = \dfrac{27}{125}$ **[M1]**

Area ratio $= \left(\sqrt[3]{\dfrac{27}{125}}\right)^2 = \left(\dfrac{3}{5}\right)^2 = \dfrac{9}{25}$ **[M1]**

Let surface areas be $9x$ and $25x$. $9x + 25x = 340 \implies 34x = 340 \implies x = 10$ **[M1]**

$25 \times 10 = 250$ **[A1]**

(b) Volume of small cone $= 1 - \dfrac{7}{8} = \dfrac{1}{8}$ of Cone B **[M1]**

Ratio of heights $= \sqrt[3]{\dfrac{1}{8}} = \dfrac{1}{2}$ **[M1]**

Height of small cone $= 20 \times \dfrac{1}{2} = 10$ **[A1]**

[Total Marks: 7]

5 **(a)** Angle AXB = Angle CXD (vertically opposite) **[B1]**

Angle ABX = Angle CDX (alternate angles, AB parallel to DC) **[B1]**

(Accept Angle BAX = Angle DCX for the second B1 mark)

(b) Length scale factor $= \frac{15}{9} = \frac{5}{3}$ **[M1]**

$DX = 6 \times \frac{5}{3} = 10$ **[M1]**

$BD = 6 + 10 = 16$ **[A1]**

(c) Area scale factor $= \left(\frac{15}{9}\right)^2 = \frac{25}{9}$ **[M1]**

Area $= 27 \times \frac{25}{9} = 75$ **[A1]**

(d) Ratio of bases $\frac{CX}{AX} = \frac{15}{9} = \frac{5}{3}$ **[M1]**

Area $= 27 \times \frac{5}{3} = 45$ **[A1]**

[Total Marks: 9]

6 **(a)** Slant height $l = \sqrt{10^2 + 24^2} = 26$ **[M1]**

Surface Area $= \pi(10)^2 + \pi(10)(26)$ **[M1]**

360π **[A1]**

(b) $V_A = \frac{1}{3}\pi(10)^2(24) = 800\pi$ **[M1]**

Volume Ratio $= \frac{100\pi}{800\pi} = \frac{1}{8}$ **[M1]**

Area Ratio $= \left(\frac{1}{8}\right)^{\frac{2}{3}} = \frac{1}{4}$ **[M1]**

Surface Area of $B = 360\pi \times \frac{1}{4} = 90\pi$ **[A1]**

(c) Volume of frustum $= 800\pi - 100\pi = 700\pi$ **[M1]**

$\frac{4}{3}\pi R^3 = 700\pi$ **[M1]**

$R^3 = 700 \times \frac{3}{4} = 525$ **[M1]**

$R = \sqrt[3]{525} = 8.07$ (accept $8.066\ldots$) **[A1]**

[Total Marks: 11]